Introduction to Relational Databases and SQL



Today's agenda



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Today's Questions

- Why use a database?
- What is a relational database?
- Why use a relational database?
- How do you use a relational database?
- What is SQL?
- Review

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Why use a database?

Why Use a Database?



Structured

Databases place *constraints* on the data they contain that ensure **data integrity**. For example, not allowing letters to be input into a numerical field like price. This minimizes errors and improves future analysis.



Non Local

Databases allow us to store vast amounts of information non-locally so that it can be accessed by many different individuals in many different locations.

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What is a relational database?

What is a Relational Database?



Related Entities

Rather than containing all the data in a single table or a set of independent tables like you might in excel or google sheets, relational databases chunk the data into separate entities with similar properties/features and then allow these separate tables to reference one another in a precise way.

Let's take a closer look at how this works…

What is a Relational Database?

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Order Number	Cost	Order Timestamp	Customer Name	Customer Phone Number	Customer Email
0001	12.50	2024-01-15T13:45:30	Joan	123456789	joan@gmail.com
0012	15	2024-01-23T12:35:30	Joan	123456789	joan@gmail.com
0043	14.50	2024-03-02T09:40:00	Joan	123456789	joan@gmail.com

Redundancy

As we can see, in a simple flat-file database like the kind you might have in excel, we can have a lot of redundant information that is wasting storage space.

What is a Relational Database?

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0043	14.50	2024-03-02T09:40:00	Joan	123456789	joan@gmail.com

Separating by Entity Type

To convert this to a relational database, we want to recognize the different entities being represented. In this case we have several features that are properties of customers, and several that are properties of orders.

What is a Relational Database?

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Order Number	Cost	Order Timestamp
0001	12.50	2024-01-15T13:45:30
0012	15	2024-01-23T12:35:30
0043	14.50	2024-03-02T09:40:00

Customer Name	Customer Phone Number	Customer Email
Joan	123456789	joan@gmail.com
Joan	123456789	joan@gmail.com
Joan	123456789	joan@gmail.com

Separating by Entity Type

Separating these features into different tables will allow us to avoid repeating information.

What is a Relational Database?

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What is a Relational Database?

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0012	15	2024-01-23T12:35:30
0043	14.50	2024-03-02T09:40:00

Primary key

Customer ID	Customer Name	Customer Phone Number	Customer Email
0013	Joan	123456789	joan@gmail.com

Connecting Tables

We still need to keep track of which customer made the order, so we have to implement some way of uniquely identifying each customer so that they can be referenced in other tables. This unique identifier is called a **primary key**.

What is a Relational Database?

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Foreign key

Order Number	Cost	Order Timestamp	Customer ID
0001	12.50	2024-01-15T13:45:30	0013
0012	15	2024-01-23T12:35:30	0013
0043	14.50	2024-03-02T09:40:00	0013

Primary key

Customer ID	Customer Name	Customer Phone Number	Customer Email
0013	Joan	123456789	joan@gmail.com

Connecting Tables

When we reference an entity in another table, we use their primary key to identify them. When a table's primary key is utilized in another table, it is called a **foreign key**.

What is a Relational Database?

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Foreign key

Order Number	Cost	Order Timestamp	Customer ID
0001	12.50	2024-01-15T13:45:30	0013
0012	15	2024-01-23T12:35:30	0013
0043	14.50	2024-03-02T09:40:00	0013

Primary key

Customer ID	Customer Name	Customer Phone Number	Customer Email
0013	Joan	123456789	joan@gmail.com

Connecting Tables

Foreign keys are essential to connect tables together and form a proper relational database.

What is a Relational Database?



Database Schema

Once we have separated our data into distinct but related entities–now represented by separate tables linked together by foreign keys–we can visualize the overall structure of the database with a chart of the various tables and their links.

This structure is called the database's **schema.**

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Why use a relational database?

Why use a Relational Database?



Data Integrity

Separating the data into different tables not only reduces **redundancy**, it also improves **data integrity**.

This is because when information is only stored once and simply referenced with foreign keys, updating it will simultaneously update it everywhere it is referenced, ensuring no inconsistencies.

Why use a Relational Database?



Queries

Relational databases also lend themselves to more complex **queries**, a concept we will explore shortly…

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How do we use a relational database?

How to Use a Relational Database

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Querying

With information spread across multiple tables, it can be hard to see what's going on when looking at any particular table.

To interact with a database, especially one that is hard to interpret on its own, we must use **queries**.

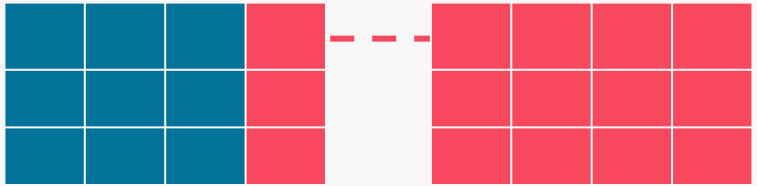


How to Use a Relational Database

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Querying

Queries provide the computer with instructions on how to combine, manipulate, and display specific desired information from a database.



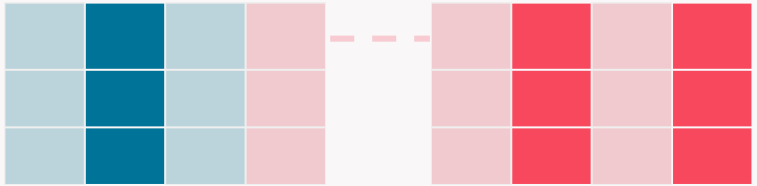
How to Use a Relational Database

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Querying

Queries provide the computer with instructions on how to combine, manipulate, and display specific desired information from a database.

The result of a query will be a table with only the information we specifically request.



How to Use a Relational Database

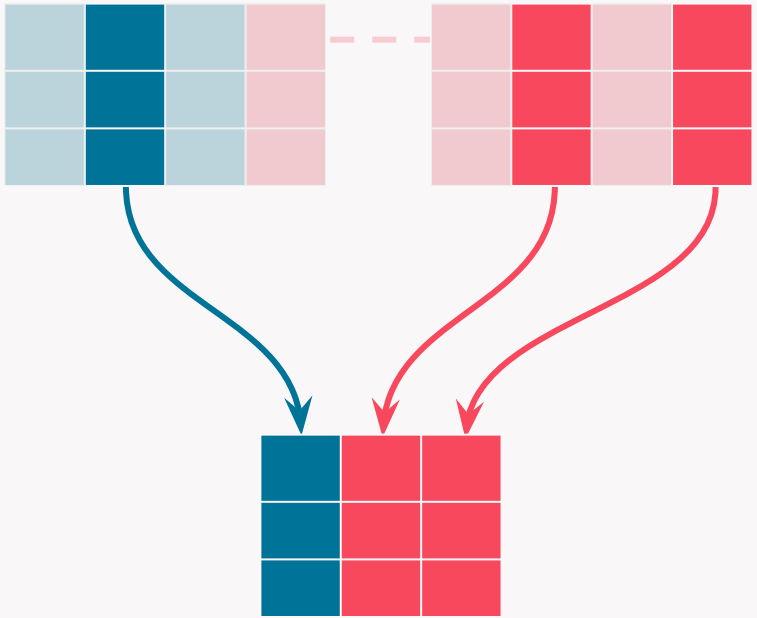
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Querying

Queries provide the computer with instructions on how to combine, manipulate, and display specific desired information from a database.

The result of a query will be a table with only the information we specifically request.

These new composite tables that queries produce are called **data marts.**



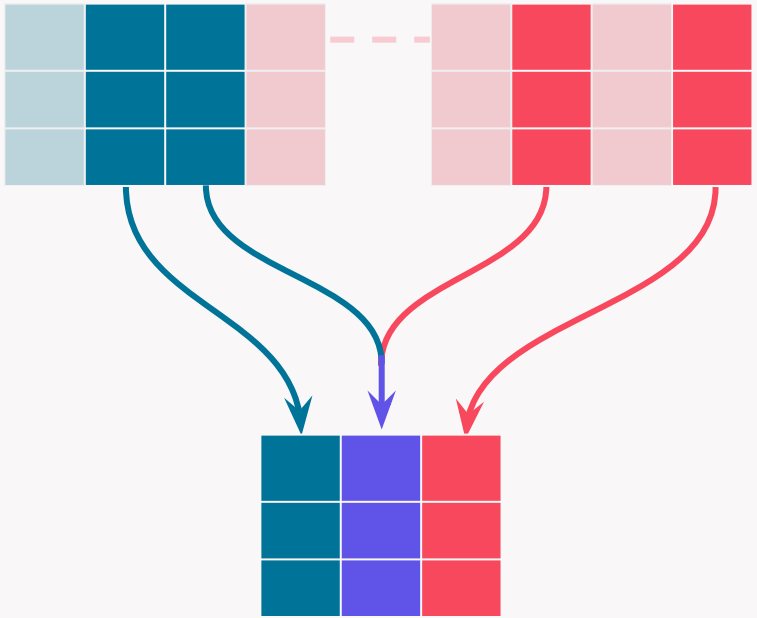
How to Use a Relational Database

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Querying

Queries may also **aggregate** or **combine** data in the database, allowing us to produce new metrics at the same time we request information.

Here we have a **combination** of two fields. This is done by combining the information from two different columns into a new metric. For instance, we might multiply price by quantity to create a new field called “total paid.”



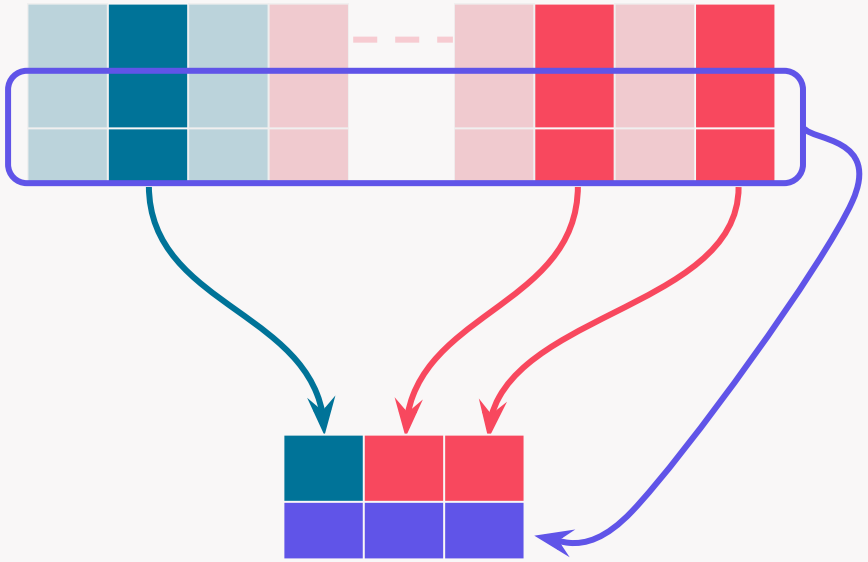
How to Use a Relational Database

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Querying

Queries may also **aggregate** or **combine** data in the database, allowing us to produce new metrics at the same time we request information.

Here we have a **aggregation** of one field. This is done by grouping multiple rows together. For instance, we might sum the “total paid” field to get the total revenue.



How to Use a Relational Database

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Querying

Complex queries can allow us to directly extract specific actionable insights from a database that are much easier to interpret than the raw data.



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What is SQL?

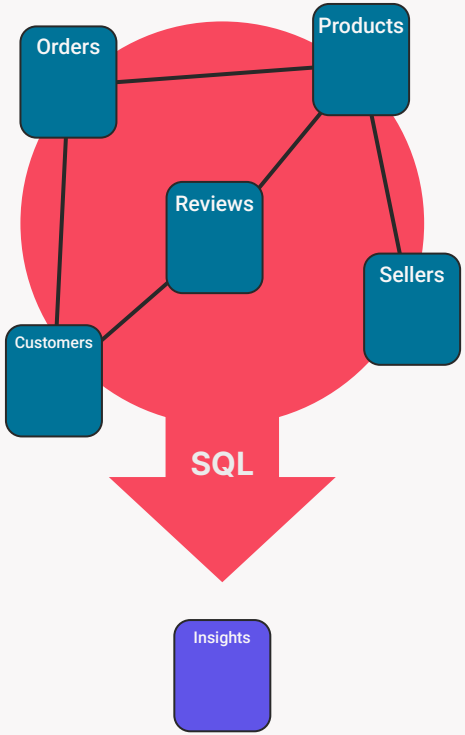
Using SQL

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Structured Query Language

SQL, or Structured Query Language, is an intuitive coding language used to query relational databases.

In other words, it is an interface for us to communicate with the database and request specific information.



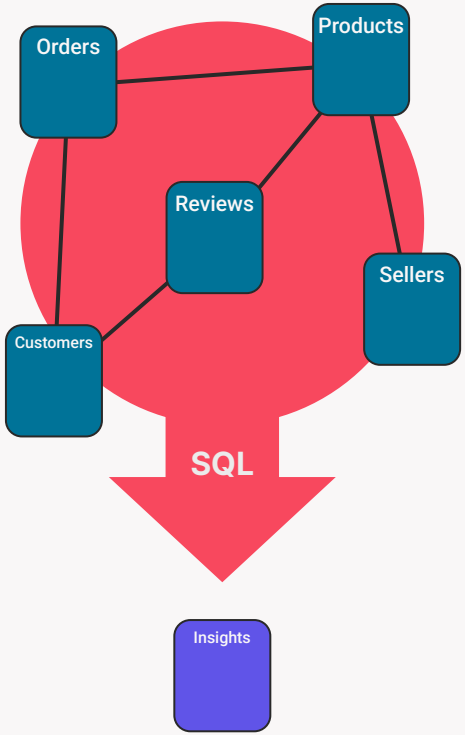
Using SQL

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Relational Database Management Systems

SQL script is written and run inside a **relational database management system** (RDBMS) where the queries can execute and retrieve the requested information.

There are a number of popular RDBMSs including PostrgreSQL, MariaDB, Oracle, IBM DB2, as well as the one we will be using, **MySQL**, which has the added benefit of being open source.



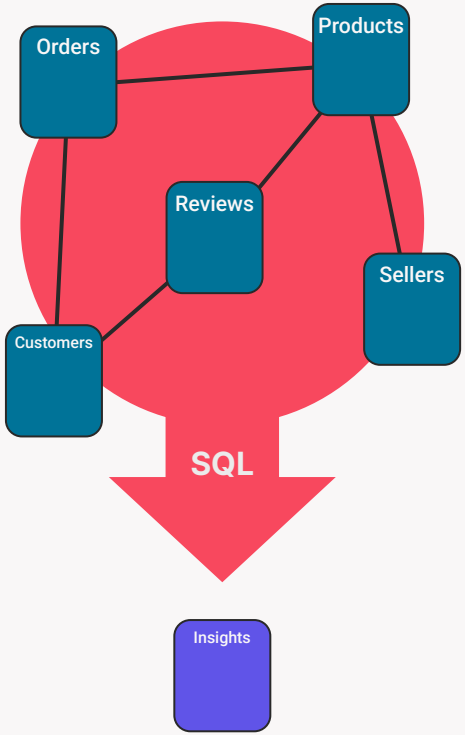
Using SQL

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SQL Syntax:

Before learning specific clauses and commands in SQL we should note some overall facts about SQL syntax:

- Queries must always end with a semicolon `;`
- Much of the formatting and indentation is flexible, but there are common practices that help make the code more readable
- SQL **keywords** are generally written in all caps



Using SQL

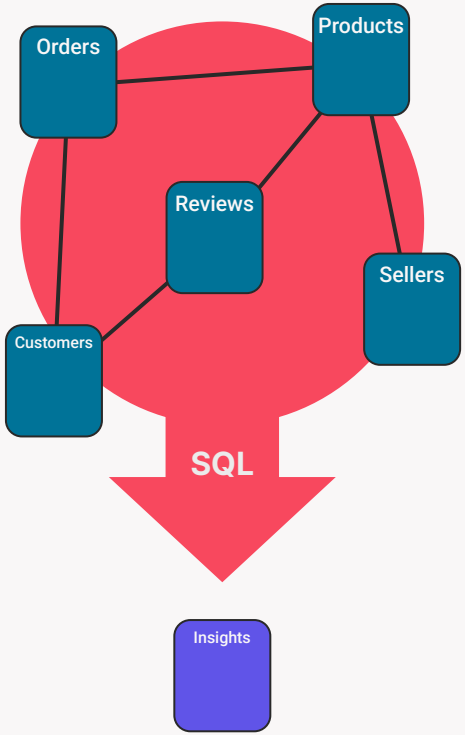
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The Basic Keywords

To start, we will learn the following essential keywords:

- `USE`
- `SELECT`
- `FROM`
- `WHERE`
- `DISTINCT`
- `ORDER BY`
- `LIMIT`
- `COUNT, SUM, AVG, MAX, MIN`
- `GROUP BY`
- `HAVING`

You can find a guide to introductory SQL here.



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Review

Review of Relational Databases and SQL

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1

Relational Databases

Relational databases split up data into separate tables to reduce redundancy and improve data integrity

2

Queries

Queries allow us to interact with the database to gather together scattered information and manipulate it into the insights we want

3

SQL

SQL is a query language. We will practice its syntax to interact with relational databases on MySQL and retrieve desired information.